

Faculty of Engineering
Electrical-Electronics Engineering Department

Power Electronics 2
Lab Experiment - Buck converter

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1. INTRODUCTION

A buck converter, also called a step-down converter, is a DC-DC converter used to reduce an input voltage to a lower regulated output voltage. It operates by switching semiconductor devices at high frequency and filtering the output with an inductor and a capacitor, which makes it more efficient than linear regulators.

In this laboratory experiment, a digitally controlled synchronous buck converter will be designed, simulated, assembled, and tested. The converter is designed to regulate a 15 V input voltage to a 5 V output voltage, with a nominal output power of 10 W and a switching frequency of 100 kHz. The design must keep the inductor current ripple below 30% and the output voltage ripple below 5%.

For an ideal buck converter operating in continuous conduction mode, the output voltage is controlled by the PWM duty cycle:

$$V_o = D V_{in}$$

where V_o is the output voltage, V_{in} is the input voltage, and D is the duty cycle.

PROCEDURE:

Simulation in PLECS:
The converter is first simulated in PLECS to verify the theoretical design and analyze voltage, current, and ripple behavior before building the real circuit.

Digital Control:
The converter is controlled using an STM32 microcontroller, which generates the PWM signals and adjusts the duty cycle according to the output voltage reference.

Real-life Assembly:
After the simulation, the circuit is assembled using the MOSFETs, gate driver, inductor, capacitor, load resistor, and microcontroller connections.

Oscilloscope Measurements:
The main waveforms are measured using an oscilloscope to verify the correct operation of the converter.

Measurements and Data Collection:
The experimental results are compared with the theoretical calculations and PLECS simulation results. The main measurements include MOSFET gate signals, switching node voltage, inductor current, output voltage, current ripple, and voltage ripple.

The objective is to verify that the converter satisfies the design requirements and maintains a stable regulated output voltage.

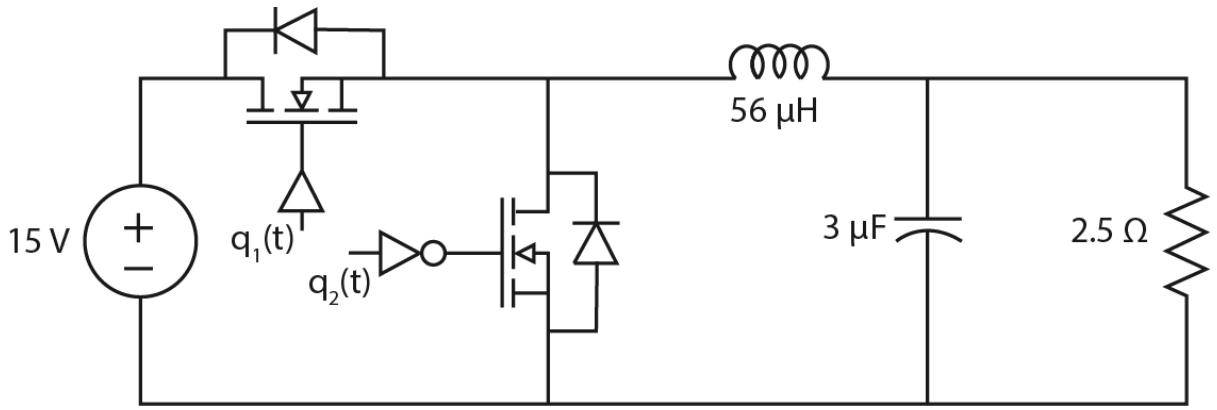
1.1 SHEET OF MATERIALS

1. STM32 microcontroller board
2. UCC27201 gate driver
3. Two MOSFETs
4. 56 µH inductor
5. 3 µF output capacitor
6. 2.5 Ω load resistor
7. 15 V DC power supply
8. Input capacitors
9. Gate resistors
10. Voltage feedback resistors
11. Ceramic decoupling capacitors
12. Perfboard or PCB
13. Connection wires
14. Oscilloscope
15. Current probe
16. Computer with PLECS software

2.ANALYSIS OF THE BUCK CONVERTER

The converter specifications are:

- Input voltage: $V_{in} = 15\,V$
- Output voltage: $V_o = 5\,V$
- Output power: $P_o = 10\,W$
- Switching frequency: $f_s = 100\,kHz$
- Current ripple: $\Delta I_L < 30\%$
- Voltage ripple: $\Delta V_o < 5\%$



The duty cycle of the converter is calculated as:

$$D = \frac{V_o}{V_{in}} = \frac{5}{15} = 0.333$$

The switching frequency of the converter is:

$$f_s = 100\,kHz$$

The switching period is calculated as:

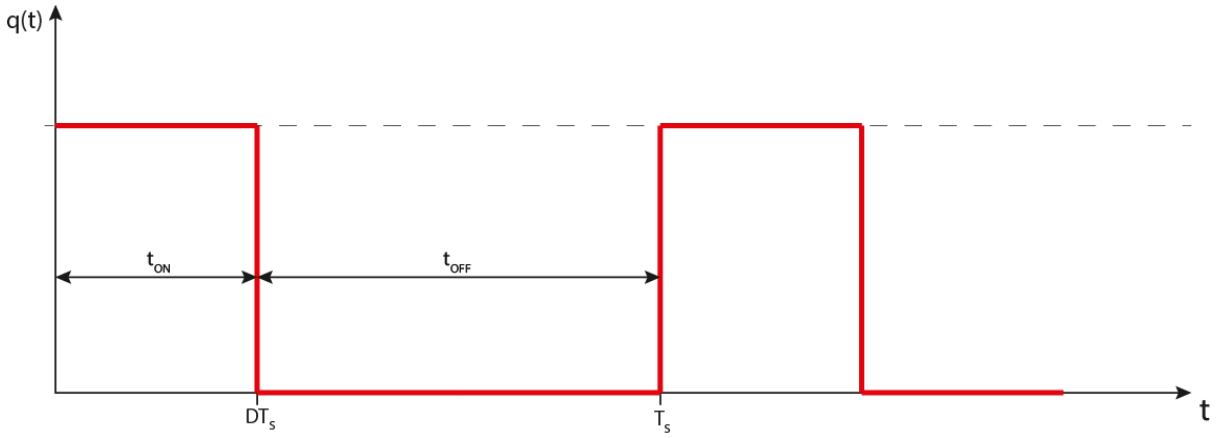
$$T_s = \frac{1}{f_s} = \frac{1}{100000} = 10\,\mu s$$

The ON and OFF times of the switch are:

$$t_{on} = DT_s = 0.333 \times 10\,\mu s = 3.33\,\mu s$$

$$t_{off} = (1 - D)T_s = (1 - 0.333) \times 10\,\mu s = 6.67\,\mu s$$

The following figure shows the switching signal of the buck converter, including the ON and OFF intervals during one switching period:



The output current is:

$$I_L = I_o = \frac{P_o}{V_o} = \frac{10}{5} = 2\,A$$

The inductor ripple current was selected as 30% of the output current:

$$\Delta I_L = 0.3 I_o = = 0.3(2) = 0.6\,A$$

The maximum and minimum inductor currents are:

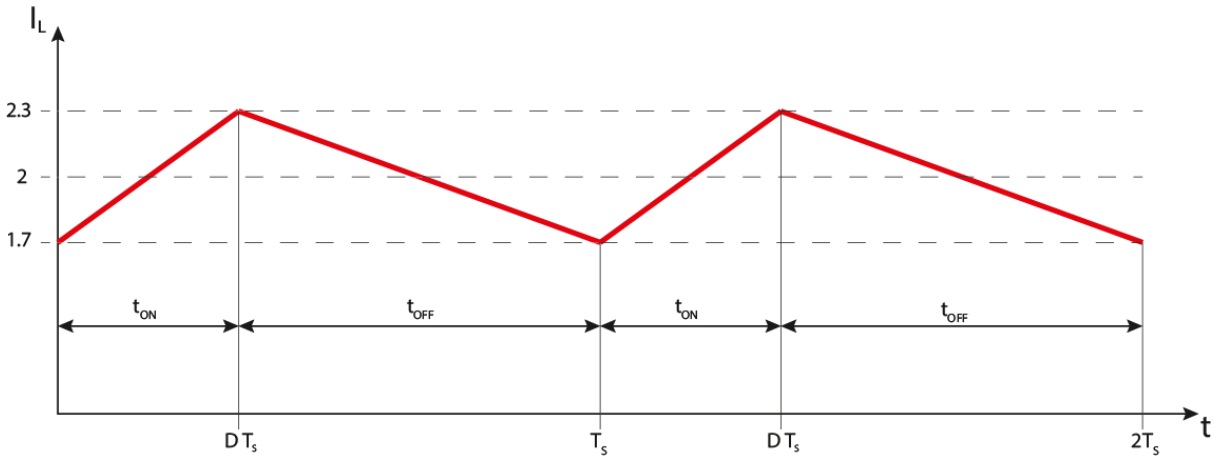
$$I_{L,\max} = I_o + \frac{\Delta I_L}{2} = 2 - 0.3 = 1.7\,A$$

$$I_{L,\min} = I_o - \frac{\Delta I_L}{2} = 2 + 0.3 = 2.3\,A$$

The required inductance value is obtained from:

$$L = \frac{V_o(1-D)}{\Delta I_L f_s} = \frac{5(1-0.333)}{0.6(100\times 10^3)} = 55{,}556\,\mu H \approx 56\,\mu H$$

The following graph illustrates the inductor current ripple and the variation between the minimum and maximum current values:



The output voltage ripple was selected as 5% of the output voltage:

$$\Delta V_o = 0.05 V_o = 0.05(5) = 0.25\,V$$

The maximum and minimum output voltage are:

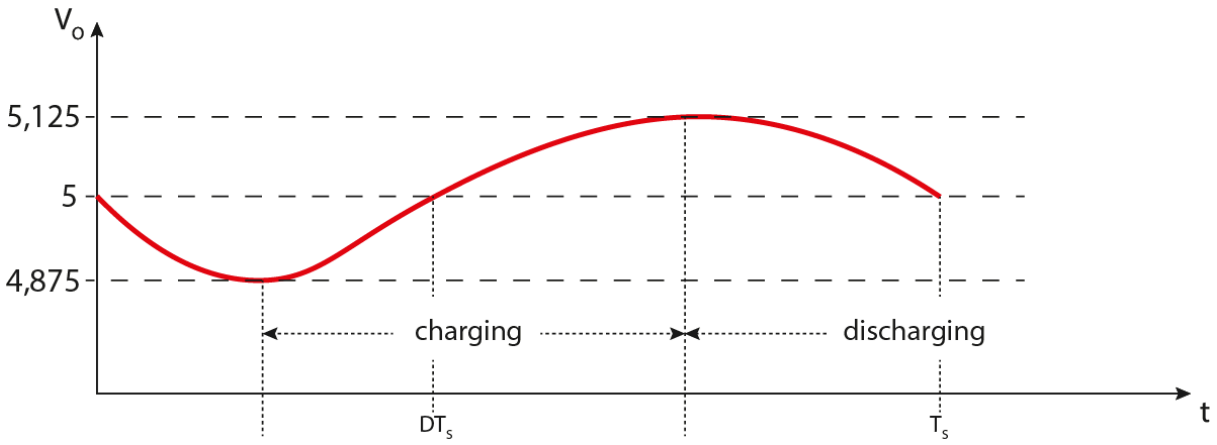
$$V_{o,\max} = V_o + \frac{\Delta V_o}{2} = 5 + 0.125 = 5.125\,V$$

$$V_{o,\min} = V_o - \frac{\Delta V_o}{2} = 5 - 0.125 = 4.875\,V$$

The minimum output capacitance is calculated using:

$$C = \frac{V_o(1-D)}{8L\Delta V_o f_s^2} = \frac{5(1-0.333)}{8(56\times10^{-6})(0.25)(100\times10^3)^2} = 2.9762\,\mu F \approx 3\,\mu F$$

Finally, the following figure presents the output voltage ripple around the average output voltage of the converter:



3.PLECS MODEL SIMULATION

Before moving to hardware prototyping, a digital simulation utilizing PLECS software was employed to validate the mathematical model. This simulation, incorporating the calculated passive components, allowed for the analysis of both the transient and steady-state behavior of the converter.

3.1. POWER STAGE AND CONTROL LOOP MODELING

The power stage (represented in Figure 1) contains several key components: a 15V DC input source, a 56µH inductor (theoretical value), a 3µF capacitor, and a 2.5Ω load resistor. This load resistor is specifically designed to dissipate the nominal 10W of power. For precise voltage regulation, even when faced with variations in the load or input line, a closed-loop control system was integrated. This system is managed by the microcontroller block.

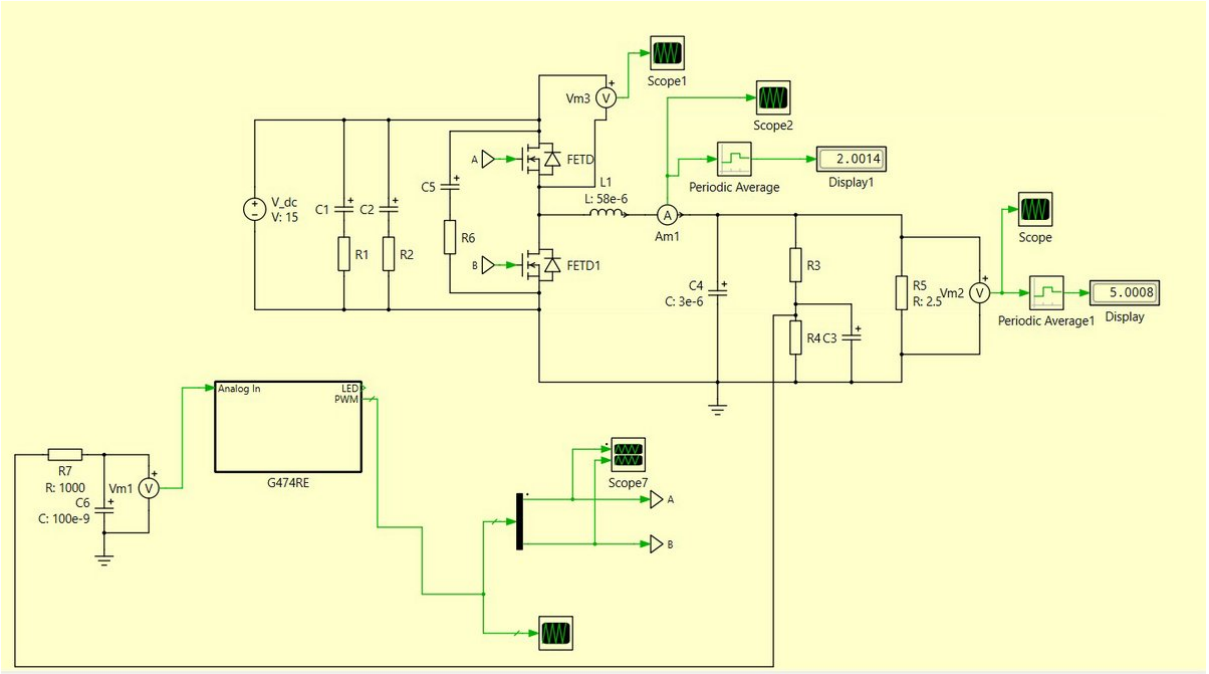


Figure 1: PLECS schematic of the synchronous buck converter

As shown in Figure 2, the control algorithm compares the measured output voltage (feedback from ADC 1) against the strict 5V reference. The resulting error signal is processed by a Proportional-Integral (PI) controller tuned with Kp=0.05and Ki=10. A saturation block limits the output duty cycle between 0 and 0.5 to guarantee system stability and prevent overmodulation before feeding the final control effort to the PWM generator.

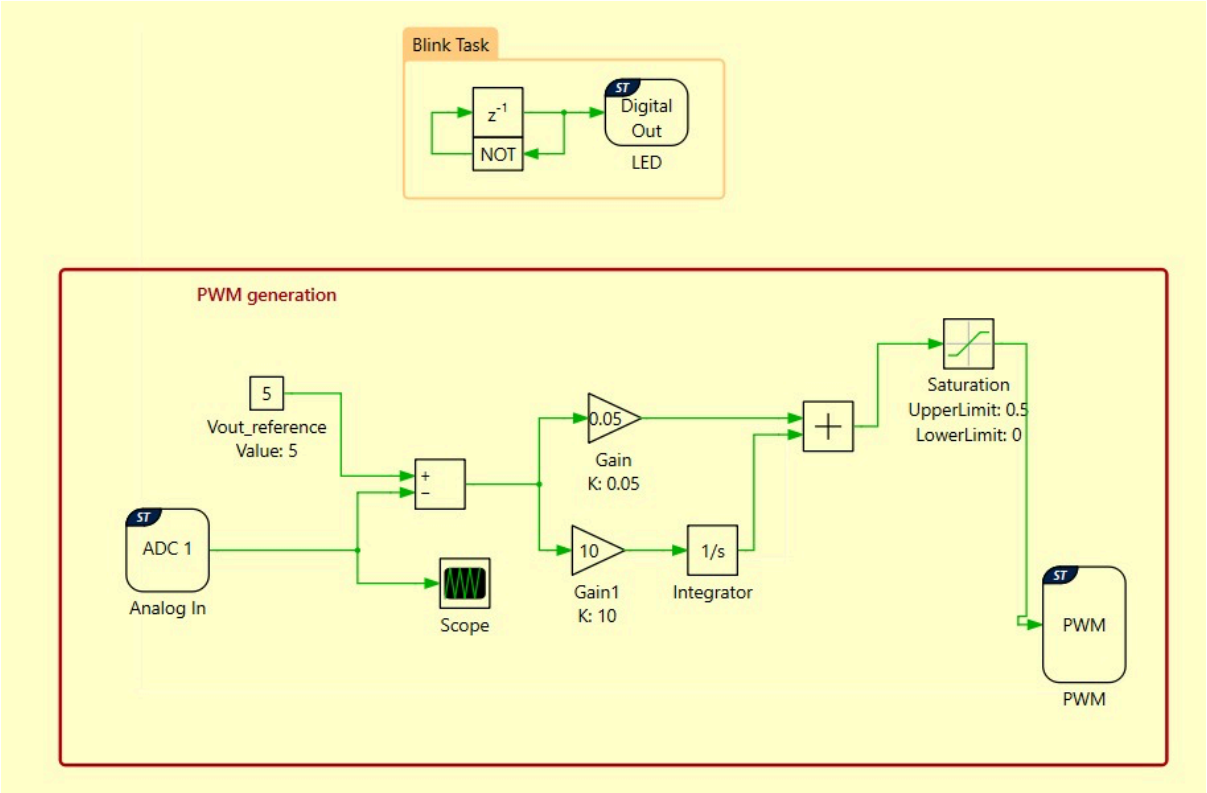


Figure 2: PI controller with closed feedback loop

2.2. SIMULATION WAVEFORMS AND RIPPLE ANALYSIS

The steady-state behavior of the converter was analyzed to verify compliance with the strict ripple constraints ($\Delta I_L \leq 30\%$ and $\Delta V_o \leq 5\%$).

The simulation cursors (Figure 3) measure a peak-to-peak inductor current ripple of 0.58A (oscillating between approximately 1.71A and 2.29A). This result perfectly complies with the theoretical maximum allowable ripple of 0.6A(30% of the 2A nominal current), confirming the correct sizing of the 56µH inductor and validating Continuous Conduction Mode (CCM).

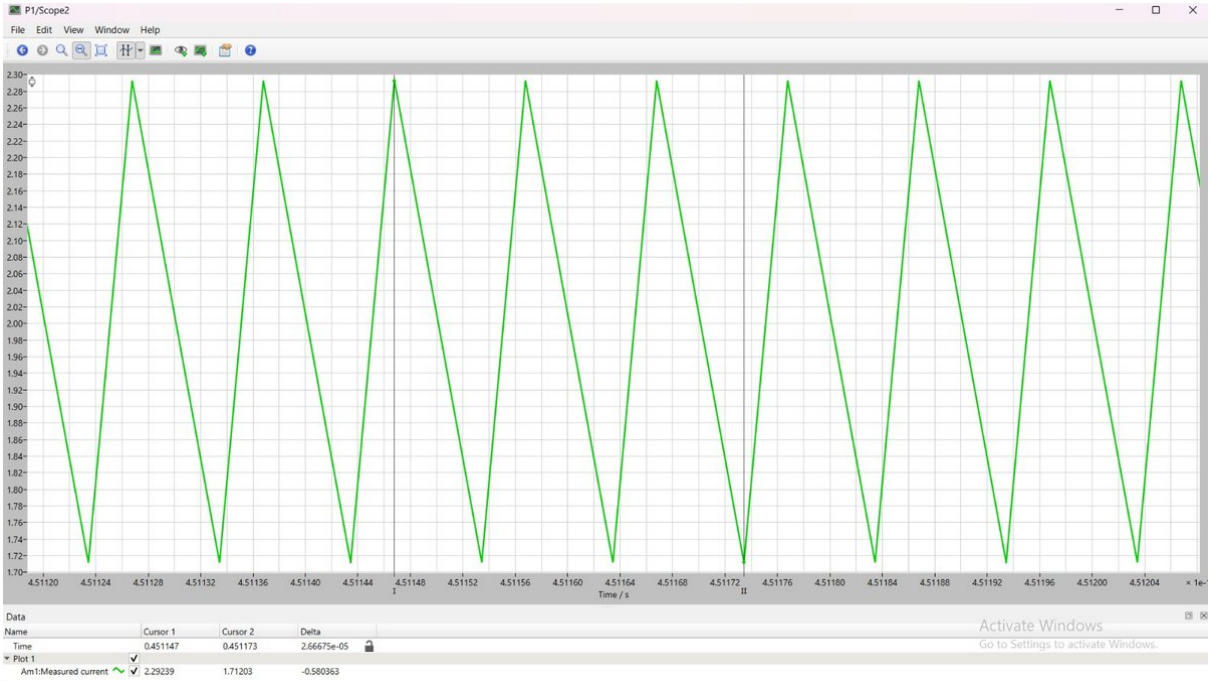


Figure 3: Inductor Current with 30% ripple

The scope in Figure 4 demonstrates a steady-state output voltage centered precisely around the 5V target, thanks to the PI controller's integral action eliminating steady-state error. The measured peak-to-peak voltage ripple is 0.235V (Delta: -0.23497V). This successfully meets the strict design constraint, remaining below the maximum 0.25V (5%) limit, thereby validating the 3µF capacitor selection.

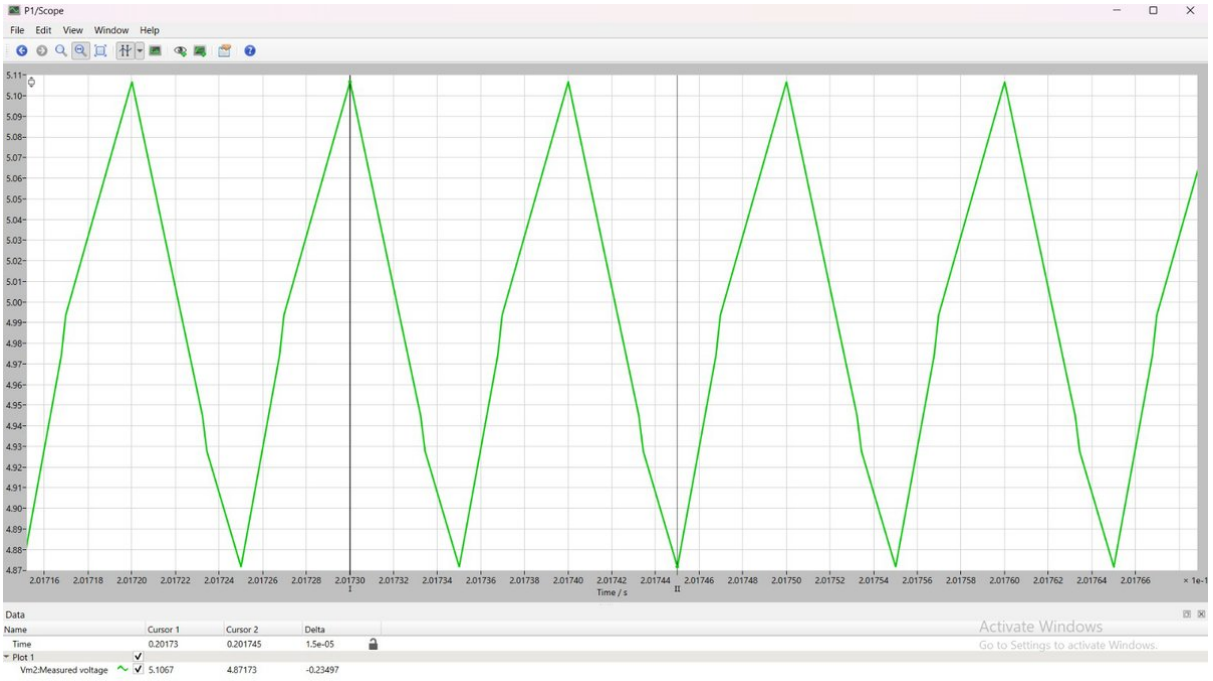


Figure 4: Output voltage with 1% ripple

Figure 5 illustrates the control signals generated for the synchronous topology. The complementary nature of the square waves ensures that the high-side and low-side MOSFETs are driven in perfect alternation, optimizing conduction efficiency and executing the synchronous rectification flawlessly.

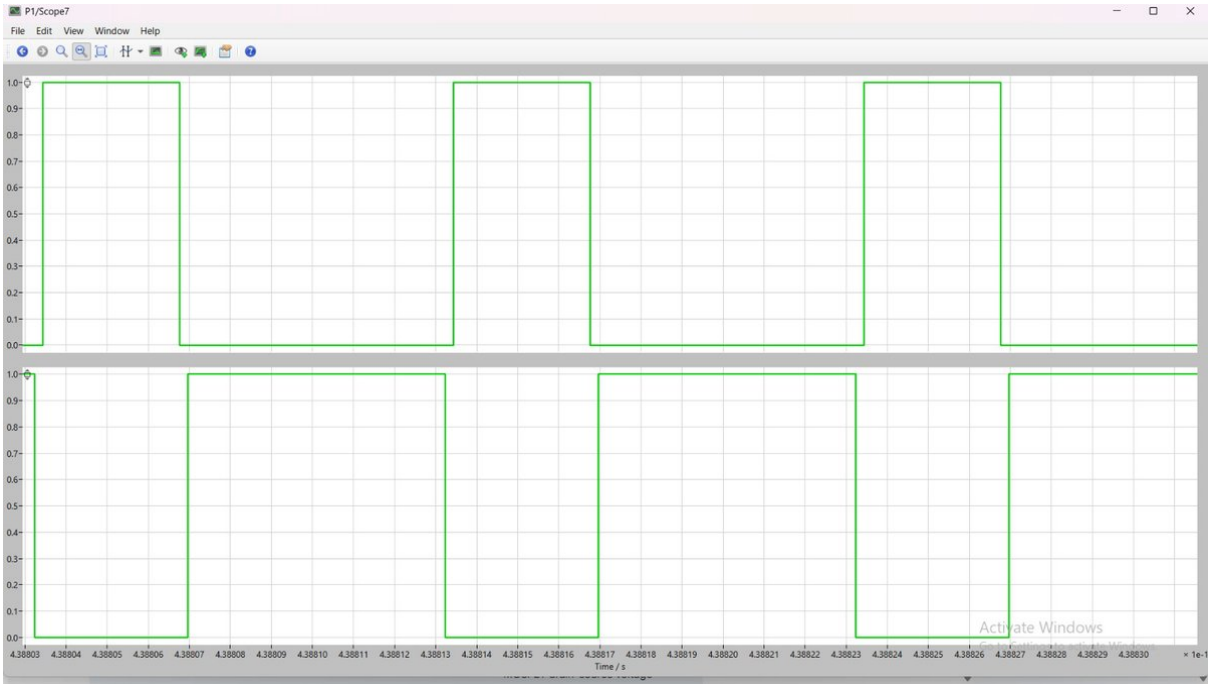


Figure 5: Switching signals - high and low signals

4. EXPERIMENTAL RESULTS

4.1.PROTOTYPE SETUP

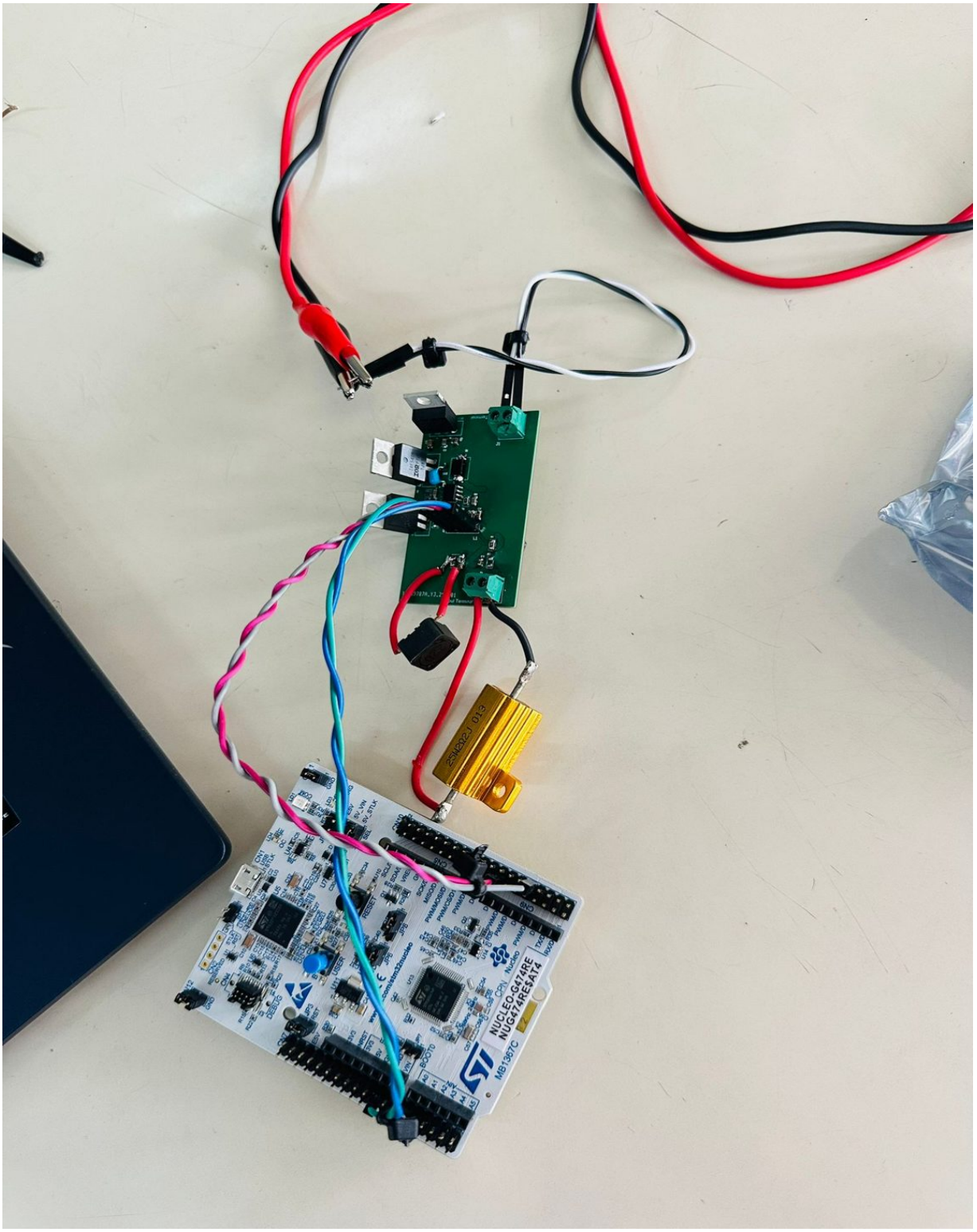


Figure 6: Physical construction of the synchronous buck converter

4.2 OSCILLOCOPE WAVEFORMS

To validate the performance of the prototype, some nodes of the circuit have been tested through an oscilloscope.

The triangular waveform (Figure 7) clearly exhibits the charge and discharge cycles of the 56µH inductor. The current consistently remains above zero, confirming Continuous Conduction Mode (CCM) operation as designed.

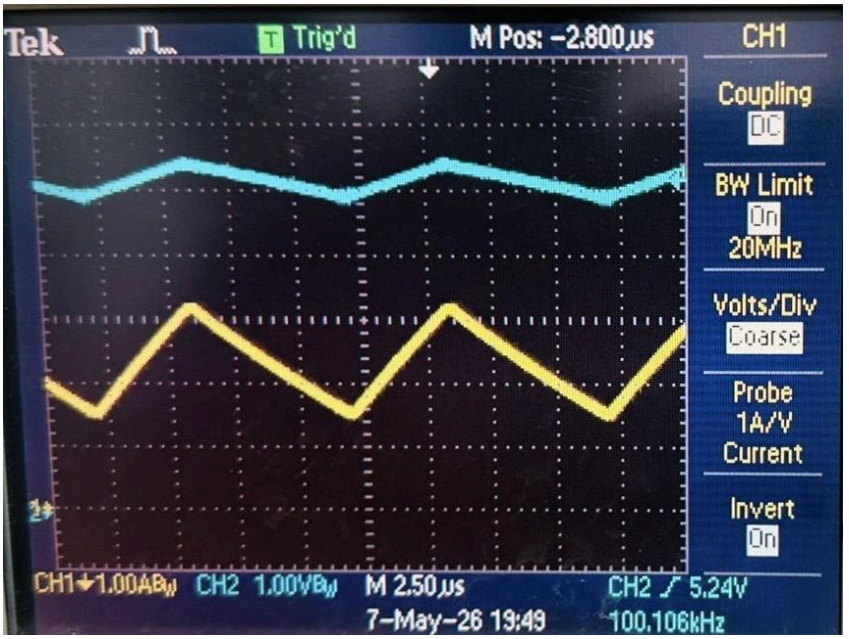


Figure 7: Experimental measurement of the inductor current waveform.

The experimental output voltage (Figure 8) stabilized at 5.24V. This introduces a minor error from the ideal 5.0V target, which is primarily attributed to non-ideal factors omitted in basic equations, such as the Direct Current Resistance (DCR) of the inductor and the RDS(on) of the conducting MOSFETs.

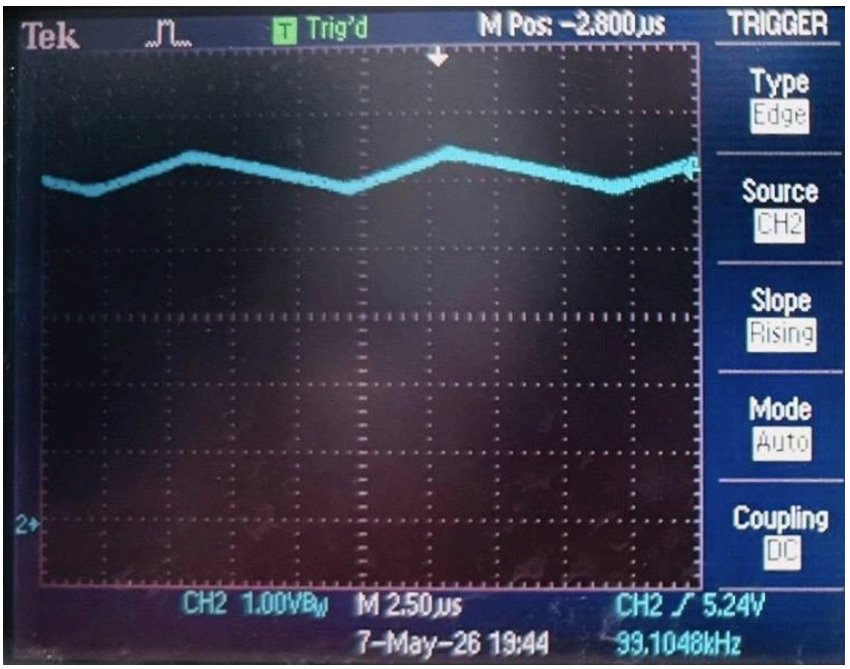


Figure 8: Steady-state output voltage measured across the load.

The oscilloscope reading shows a clean square wave (Figure 9) with an accurate frequency of ~99.1 kHz, exceptionally close to the 100 kHz design requirement. The visual duty cycle accurately reflects the theoretical 33.3%.

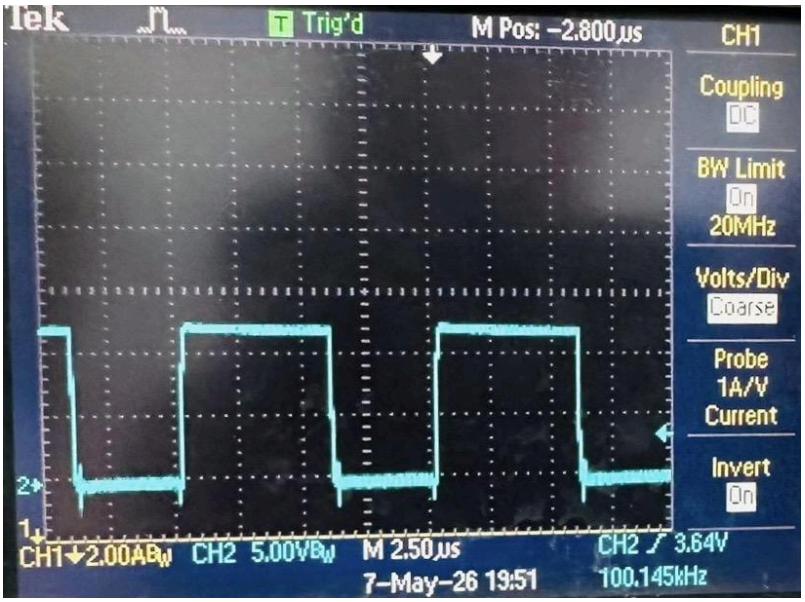


Figure 9: PWM gate pulse generated by the STM32 microcontroller.

The VDS waveform (Figure 10) cleanly transitions between 0V (conduction) and the 15V input voltage boundary (blocking state). The presence of minimal ringing implies a good PCB layout and validates the proper implementation of dead-time by the STM32 to prevent shoot-through currents.

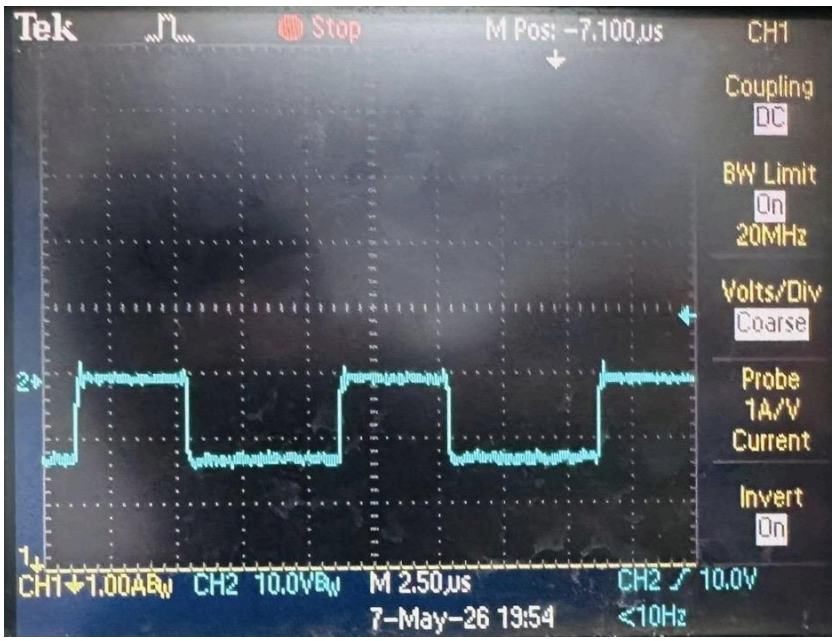


Figure 10: MOSFET Drain-Source voltage (VDS) at the switching node.

5. CONCLUSION

The synchronous buck converter project successfully attained all delineated technical and operational objectives. The design focused on the conversion of a nominal input voltage of 15V to a regulated 5V output, specified for a 10W load. Based upon theoretical analysis, the requisite LC filter parameters were established as a 56µH inductor and a 3µF capacitor. These computations specifically ensured sustained operation within the Continuous Conduction Mode (CCM) and maintained ripple levels substantially below the imposed thresholds (less than 30% for inductor current and 5% for output voltage).

The simulation results confirmed compliance with the design ripple constraints, showing an inductor current ripple close to the theoretical 30% limit and an output voltage ripple below the maximum allowable 5%. The implemented PI controller provided stable closed-loop regulation, maintaining the output voltage centered around the desired 5V reference.

Experimental validation substantiated the design's robustness. Oscilloscope measurements registered a stable output voltage of 5.24V, which corresponds to a precision error of 4.8% relative to the theoretical target of 5.0V. Moreover, the STM32 microcontroller generated a high-precision switching signal, with a measured frequency of 100.106 kHz, thereby strictly adhering to the design specification of 100 kHz.

The slight deviation in the output voltage (+0.24V) is standard in physical hardware implementations and can be attributed to real-world non-idealities omitted from ideal theoretical models, such as the inductor's Equivalent Series Resistance (ESR), the RDS(on) of the IRF540N MOSFETs, and minor voltage drops across the perfboard traces. Ultimately, the strong correlation between the analytical calculations, PLECS simulations, and physical laboratory measurements proves the converter's ability to perform highly efficient and stable power transfer under a 2.5 Ω equivalent load, fully validating the power electronics design process.

Overall, the agreement between theoretical analysis, simulation results, and experimental measurements demonstrates the successful implementation of the synchronous buck converter and confirms the effectiveness of the proposed control system and hardware design.